Mohammed Fayas

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Embedded Engineer|Robotics & Interactive Techology

Embedded Systems and Robotics Engineer with 3+ years of professional experience, specializing in robotics, interactive event technologies, and embedded systems development. Skilled in robotic arm programming and choreography, sensor integration, motor control, development boards (Arduino, ESP32, and Raspberry Pi), Embedded C and Python programming, hardware integration, and real-time interactive systems. Experienced in developing, testing, and troubleshooting embedded and automation solutions for technology-driven projects and interactive event environments. Passionate about innovation, AI, robotics, and emerging technologies.

SKILLS

- Embedded Systems Development (Arduino, ESP32/ESP8266) | Raspberry Pi Development & Integration.
- Embedded C / C++ Programming | Python for Robotics & Automation.
- Robotics & Simulation | ROS 2 (Humble), SLAM Toolbox, Navigation2 (Nav2), RViz2 Visualization, Gazebo Simulation, Robot Modeling, Mapping, and Autonomous Navigation.
- Robotics Systems Integration | Robotic Arm Control & Programming (KUKA,Kawasaki, UR3,Dobot).
- Computer vision & AI (OpenCV,YOLO,MediaPipe) | Sensor Integration (IR, Ultrasonic, Proximity, LiDAR, IMU,F/T sensors,Touch sensors,) | Depth Cams |.
- Servo Motor Control (AC/DC Servo Systems) ,Motor Drivers Actuators & Motion Control Systems.
- Serial and Industrial communication protocols including UART, I2C, SPI, CAN, RS-485, and Ethernet.
- Hardware–Software Integration & System Testing | Circuit Debugging & Troubleshooting.
- Interactive Technologies (DMX,OSC,Resolume,LED Displays) Installation, Setup & Commissioning.

WORK EXPERIENCE

Lazulite Technology Services • Umm-Al Quwain • 06/2024 - Present

Embedded Systems Engineer • Full-time

- Developed and integrated embedded systems and robotics solutions for interactive events and automation projects.
- Programmed and configured microcontrollers, including Arduino and ESP32 for real-time control applications.
- Integrated sensors, motors, actuators, and robotic arm systems for interactive installations and automation setups.
- Programmed industrial robotic arms and developed synchronised motion sequenses and choreography for live events.
- Designed and implemented hardware-software communication using Python and embedded platforms.
- Performed hardware integration, system prototyping, testing, and troubleshooting for deployed projects.
- Assisted in the setup, calibration, and maintenance of interactive technology systems during exhibitions and live events.
- Collaborated with technical and creative teams to deliver immersive automation experiences across UAE-based projects.
- Conducted system diagnostics, performance optimization, and on-site technical support during deployments.
- Managed real-time control systems and ensured reliable operation of embedded and robotic solutions.

Indian Naval Academy • **Kannur** • kerala• **09/2023 - 06/2024**
Assistant Engineer • **Full-time**

- Performed maintenance and technical support for advanced training laboratory equipment at the Indian Naval Academy.
- Serviced and troubleshot radar systems, microwave laboratory equipment, and satellite communication trainer kits.
- Maintained and calibrated sensor kits, machine laboratory equipment, and AC/DC servo systems.
- Assisted in the installation, testing, and operational support of electronic and electromechanical training systems.
- Provided hands-on technical support for student training projects and laboratory practical sessions.
- Conducted diagnostics, preventive maintenance, and fault rectification to ensure reliable system performance.
- Supported procurement activities for new laboratory equipment, components, and technical resources.
- Coordinated project kit preparation, delivery, and setup for academic and technical training activities.
- Ensured proper documentation, inventory management, and operational readiness of laboratory systems.
- Worked in a structured defense training environment with a focus on system reliability, safety, and operational efficiency.

EDUCATION

B -Tech in Electrical and Electronics Engineering with Minor Degree in Robotics and Automation

APJ Abdul Kalam Technological University • Palakkad • GPA: 2.9 • 08/2020 - 07/2023

Diploma in Electrical and Electronics Engineering

Malabar Polytechnic Campus • Palakkad • GPA: 3.4 • 08/2017 - 09/2020

LANGUAGES

- English
- Hindi
- Malayalam (Native)